Mathieu1D Direct Finite Reduction

Calculation-facing technical report

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Status. This report describes one retained, unsealed, illustrative finite numerical execution. Independent verification checks the exact result bytes and owns the authenticated plot-data projection. Neither the calculation nor this report establishes continuum convergence, uncertainty quantification, material validation, publication acceptance, or a sealed evidence package.

Abstract

This report records a direct typed calculation of a one-dimensional Mathieu Hamiltonian, independent reconstruction of its finite numerical channels, and two figures rendered only from authenticated data. The calculation compares finite plane-wave and finite-difference representations, transforms one sampled band to a complete finite hopping inventory, and evaluates declared symmetric hopping-range truncations on separate training and evaluation coordinates. The results concern the specified finite representations and do not promote adjacent-representation changes to continuum errors or uncertainties.

1 Scientific question

For the represented dimensionless Hamiltonian

$$H = -\frac{d^2}{dx^2} + 0.5\cos x, \qquad x \sim x + 2\pi, \tag{1}$$

how do the declared finite parent representations, the complete 64-point discrete Fourier transform, and the finite symmetric hopping ranges behave on their fixed coordinate inventories?

The calculation uses plane-wave cutoffs $3, 5, 7, 9, 11$ with cutoff 15 as a finite represented reference. A separate periodic finite-difference route uses $31, 63, 127, 255$ points. The lowest of three compared finite bands is sampled on a 64-point centered training mesh and a disjoint 257-point staggered evaluation mesh. The declared maximum hopping ranges are $0, 1, 2, 3, 4, 6, 8$. All energies are nondimensional in the declared recoil-energy convention.

2 Ownership and verification boundary

Reusable source under `ksdft2effmass.model_systems.mathieu1d` owns typed configuration, calculation, result, verification, authenticated projection, and rendering behavior. The direct calculation script composes existing plane-wave, periodic finite-difference, Fourier transform, truncation, interpolation, diagnostic, and weighted least-squares owners.

The independent verifier does not call the producer analyzer. It assembles dense plane-wave fibers and sparse twisted finite-difference fibers directly, evaluates the finite Fourier sums explicitly, and

performs the weighted least-squares calculation through a separate NumPy route. It hashes the exact admitted result bytes and binds the resulting digest to the plot-data projection. Rendering reruns that verifier and requires exact equality of both the verification aggregate and the authenticated projection before publishing any figure.

3 Complete authenticated view

Figure 1 displays four finite-data channels: represented parent refinement, the sampled evaluation band and selected finite-range reconstructions, the complete finite-transform hopping magnitudes, and the declared range diagnostics. Connected marks follow explicitly ordered finite inventories. They do not assert interpolation or a continuum limit. Symmetric-logarithmic axes retain recorded zeros without replacing them by artificial positive floors.

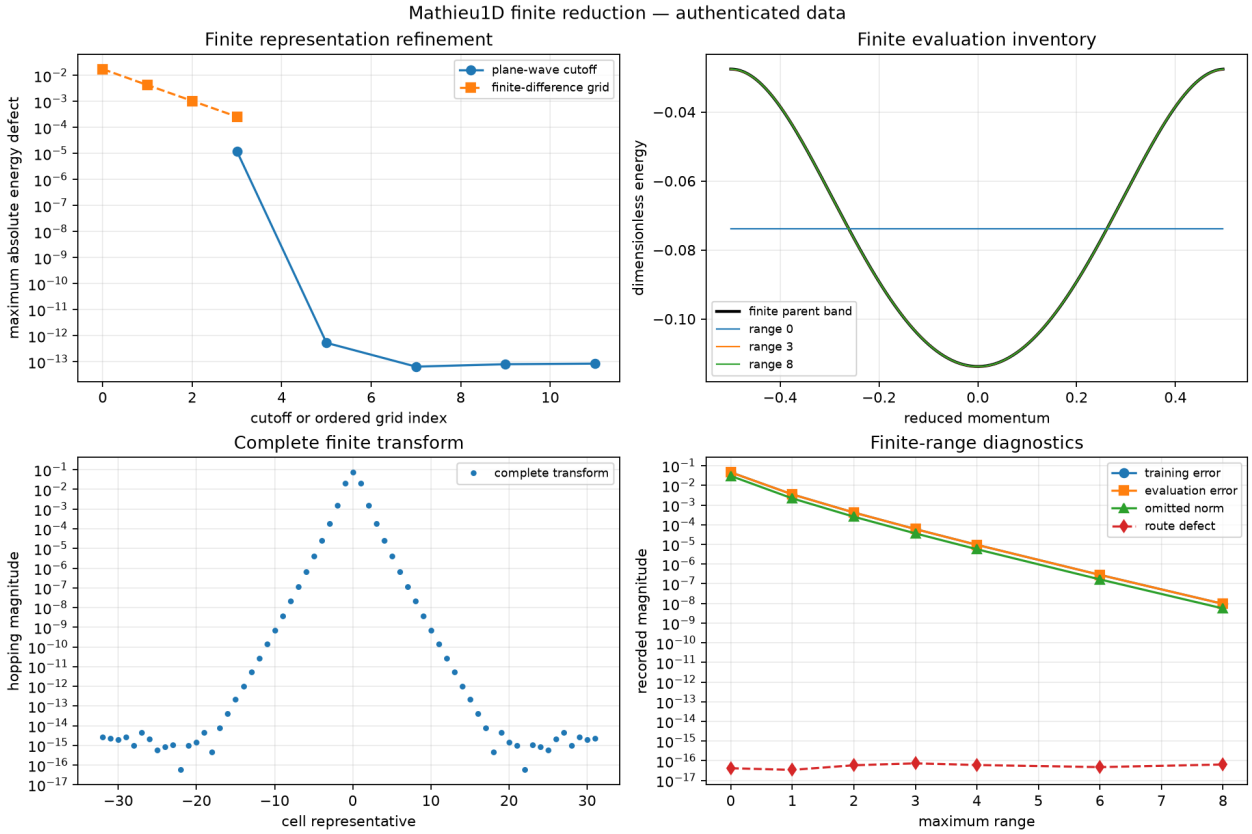


Figure 1: Complete direct Mathieu1D visualization rendered from the independently verified typed projection.

4 Publication-facing finite-range summary

Figure 2 is a compact projection of the same authenticated data. Its left panel records training and disjoint-evaluation maximum errors for every declared range. Its right panel records the corresponding omitted-block norm. The compact layout creates no new scientific values and is not an additional calculation.

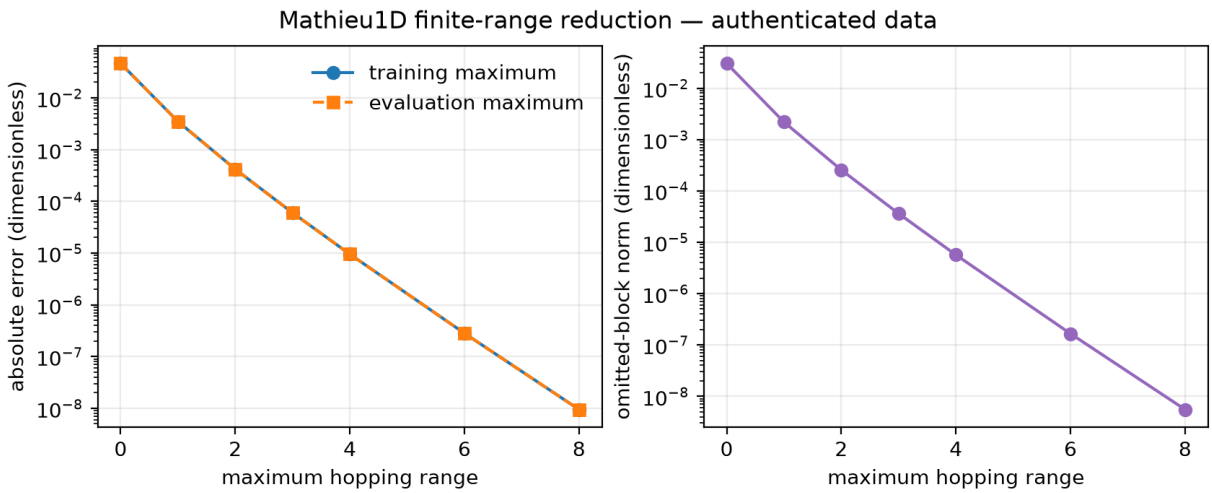


Figure 2: Compact finite-range summary derived from the same authenticated projection as Figure 1.

5 Interpretation limits

The plane-wave reference is itself finite. The finite-difference sequence is a comparison among represented matrices, not a certified continuum enclosure. The complete discrete transform reconstructs one finite sampled inventory; truncating it tests declared finite model classes. Agreement between the mediated and direct-fit routes is meaningful only under the matched coordinates, weights, and model class used here. Sampled band gaps do not certify continuum spectral isolation, and no hopping range is accepted as universally adequate.

The present direct execution is the sole live Mathieu1D calculation and publication source. Predecessor code and run artifacts are absent from the live tree and are not runtime, verification, rendering, or report inputs.